



GRETCHEN WHITMER
GOVERNOR

STATE OF MICHIGAN
DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY
WATER RESOURCES DIVISION



PHILLIP D. ROOS
DIRECTOR

September 16, 2025

VIA EMAIL

Jeffrey Dietlin
200 North Lake Street
Cadillac, Michigan 49601

Dear Jeffrey Dietlin:

SUBJECT: Biosolids Audit
National Pollutant Discharge Elimination System (NPDES)
Individual Permit No. MI0020257
Designated Name: Cadillac Wastewater Treatment Plant (WWTP)

On April 29, 2025, staff of the Department of Environment, Great Lakes, and Energy (EGLE), Water Resources Division (WRD), conducted a Biosolids Audit at the Cadillac WWTP located at 1121 Plett Road Cadillac, Wexford County, Michigan 49601. The purpose of the audit was to evaluate the facility's compliance with the conditions of your approved Residuals Management Program (RMP) and compliance with requirements contained in the Michigan Part 24 Administrative Rules, Land Application of Biosolids, of Part 31, Water Resources Protection, of the Natural Resources and Environmental Protection Act, 1994 PA 451, as amended.

Patrick Miller with the city of Cadillac and Victoria Starry with EGLE WRD participated in the audit. The audit included a discussion of operations, review of biosolid records, a facility tour, and an inspection at a biosolids site during land application. I would like to thank you for your cooperation and assistance during the audit.

The following is a summary of the audit results:

1. Recordkeeping and Reporting (R 323.2413/R 323.2416)

Records documenting land application activity, analytical data, and unit process information were available upon request and records appear to be kept at a minimum of five years as required. There are several sites subject to Cumulative Pollutant Loading Rate (CPLR) requirements due to previous copper and zinc concentration exceedances, and all required CPLR site documentation was present.

2. Sampling and Analysis (R 323.2406)

It was indicated by the facility staff that prior to collecting biosolids samples for analysis, the storage tanks are aerated for approximately 2 to 3 days to ensure a uniform material. Multiple samples are then collected from different areas within the storage tank with a sludge judge device. The samples are combined, and a composite sample is taken from the mixture.

The most recent analytical results were reviewed and are well below the Table 3 Pollutant Concentrations and the Table 1 Ceiling Concentration limits.

3. Operational Standards (R 323.2411)

Wasted solids collected from the activated sludge process are sent to the primary anaerobic digester. Digester temperatures average 98 degrees Fahrenheit with an average cell residence time of 20 days. After digestion, solids are pumped to a secondary digester and then to one of two sludge storage tanks.

Although Cadillac WWTP also complies with the Class B pathogen reduction requirements for anaerobic digestion, they certify and demonstrate Class A pathogen reduction requirements through sampling for fecal coliform, enteric virus, salmonella, and helminth ova. The analytical data demonstrating compliance with Class A pathogen reduction requirements was available on the day of the audit. The facility meets vector attraction reduction requirements through subsurface injection.

4. Management Practices (R 323.2410)

Cadillac WWTP utilizes Biotech Agronomics for their site selection, soil testing, and land application of their biosolids. A site inspection was completed in conjunction with the audit to observe the land application of biosolids by Biotech. Biosolids were transported with trucks and land applied via injection at site 21N09W25-AB01. There were no violations of isolation distances and no pooling of biosolid was observed at the time of the site inspection. No significant issues were identified during the site inspection portion of the audit.

5. Residuals Management Program (R 323.2403(3)(c))

The RMP was approved on May 6, 2002. It appears that Cadillac WWTP is implementing the conditions of the approved RMP. Any changes to the treatment or land application of the biosolids will require a modification to the facility RMP that will need to be submitted to MiEnvirofor approval.

6. Annual Report (R 323.2416(3))

The past five years of Biosolids Annual Reports and Annual Biosolids DMRs were submitted to EGLE via MiEnviro as required by the Part 24 Rules and appear to be administratively complete.

7. PFAS Monitoring Report

The most recent Biosolids PFAS Monitoring Report was submitted to MiEnviro over 2 weeks prior to beginning land application, as required. The facility continues to maintain PFOS and PFOA concentrations of less than 20 ug/kg when combined in accordance with their Class AEQ status, and all PFAS test results were available to review at the time of the audit.

We appreciate your efforts to maintain compliance with the conditions of your approved RMP and the requirements contained in the Part 24 Rules. If you have any questions or comments regarding this letter, please contact me 989-355-4805; StarryV@Michigan.gov; or at EGLE, WRD Cadillac District Office, located at P.O. Box 30458, Lansing, Michigan 48909-7958

Sincerely,

Victoria Starry
Cadillac District Office
Water Resources Division

vs/sh

cc: Patrick Miller, City of Cadillac WWTP